

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
WINTER SEMESTER EXAMINATION – 2024

Course: B. Pharmacy

Subject Name: Biopharmaceutics and Pharmacokinetics

Max Marks: 75

Date: 26/12/2024

Semester: VI

Subject Code: BP604T

Duration: 3 Hr.

LIBRARY

Instructions to the Students:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q.1. Objective Type Questions (Answer All the Questions)

(10 X 2) = 20

- i) Define Apparent volume of distribution.
- ii) Write about objectives of bioavailability studies.
- iii) Differentiate Passive transport and Active transport.
- iv) Write clinical significance of protein binding of drugs.
- v) What is BCS classification of drugs?
- vi) Write in short about non renal route of Excretion.
- vii) What are the different methods used to calculate the AUC?
- viii) Plot plasma drug concentration Vs time profile of drug.
- ix) Write formulas for calculating loading dose and maintenance dose.
- x) Enlist various factors which affect renal excretion of drugs.

Q.2. Long Answers (Answer 2 out of 3)

(10 X 2) = 20

- i) Define Non-linear pharmacokinetic. Explain methods to determine non linearity and causes of non-linearity.
- ii) Define drug absorption. Explain factors influencing absorption of drug.
- iii) Discuss in detail one compartment open model for a drug administered as IV bolus.

Q.3. Short Answers (Answer 7 out of 9)

(5 X 7) = 35

- i) Explain drug plasma-level time curve along with its Pharmacokinetics parameters.
- ii) Write a note on IVIVC correlations.
- iii) Describe about the physiological barriers to the distribution of drugs.
- iv) Explain the factors affecting protein binding of drugs.
- v) Write the methods to enhance the bioavailability of poorly soluble drugs.
- vi) Write a note on Mammillary and Catenary models.
- vii) Explain the dissolution apparatus type-I and type-II as per USP.
- viii) Explain different mechanisms of drug absorption.
- ix) Explain Phase-I reactions of drug metabolism.

*** END OF THE PAPER ***